

SmartTouch Mouse: A Trackpad-Enabled Mouse with Gesture-Controlled Multi-Device Switching

Ronnie Bas
rongreat4@gmail.com
Cavite, Philippines

Jhasper Corpuz
jhaspercorpuz25@gmail.com
Caloocan, Philippines

John Michael Llantino
llantinojohn@gmail.com
Antipolo City, Philippines

Hero James Oligane
herojamesoligane22@gmail.com
Valenzuela City, Philippines

Gian Carlo Onia
gc.g.onia@gmail.com
Quezon City, Philippines

Andrew Rubio
andrubio303@gmail.com
Manila, Philippines

Abstract—*The SmartTouch ClickPad Mouse introduces a design that replaces traditional left/right buttons with a full-width capacitive touchpad supporting multi-touch gestures. It enables seamless gesture-based multi-device switching without physical toggles, improving efficiency for educators, remote workers, designers, and power users. This innovation enhances workflow productivity, minimizes hardware clutter, and provides a more intuitive user experience with integrated haptic feedback, gesture zones, and customizable software profiles. The proposed architecture utilizes an Arduino Pro Micro as the main HID controller, which interfaces with an ESP32 for touch processing and a standard mouse PCB for optical sensing.*

Keywords—*Capacitive Trackpad, Multi-Device, Gesture Control, Human Interface Device (HID), ESP32, Arduino Pro Micro, Smart Mouse*

I. INTRODUCTION

The SmartTouch ClickPad Mouse introduces a design that replaces traditional left/right buttons with a full-width capacitive touchpad supporting multi-touch gestures. It enables seamless gesture-based multi-device switching without physical toggles, improving efficiency for educators, remote workers, designers, and power users.

This innovation enhances workflow productivity, minimizes hardware clutter, and provides a more intuitive user experience with integrated haptic

feedback, gesture zones, and customizable software profiles.

II. DESCRIPTION OF THE PROBLEM

The modern computational workflow is often fragmented, with users operating across multiple devices (desktops, laptops, tablets) [cite: 28]. This multi-device environment creates significant workflow "friction," leading to inefficiencies and challenges for various user groups. The key problems addressed by this project include:

- **Hardware Clutter and Workflow Interruption:** Power users, such as designers and remote workers, often accumulate multiple physical mice to control different systems. This leads to desk clutter and requires users to physically move their hands between devices, interrupting their focus and workflow [cite: 29].
- **Limited Input "Vocabulary":** Traditional mice offer a restricted set of inputs (left-click, right-click, scroll). This limited "vocabulary" is insufficient for the complex, gesture-based interactions common in modern operating systems and sophisticated software applications (e.g., CAD, video editing, virtual desktops). Users are forced to rely on less intuitive keyboard shortcuts, slowing down their work [cite: 2].
- **Inelegant Device Switching:** Existing solutions for multi-device control are often cumbersome. Physical KVM (Keyboard, Video, Mouse) switches require manual button presses and add to desk clutter. Software-only solutions (e.g.,

Synergy, Logitech Flow) can introduce noticeable latency, rely on network connectivity, and may not always provide a seamless transition between devices [cite: 2, 6, 31].

These problems collectively contribute to reduced productivity and a less intuitive, more disjointed user experience. The SmartTouch ClickPad aims to mitigate these issues by providing a unified, gesture-driven input device.

III. DESCRIPTION OF THE PROPOSED IMPROVEMENT

Overview

The SmartTouch ClickPad Mouse converts a salvaged laptop touchpad (Synaptics T1320A) into a universal pointing device. An ESP32 reads raw finger data through the I²C interface, processes gestures (move, click, scroll, swipe), and communicates with an Arduino Pro Micro via serial. The Pro Micro acts as a USB HID mouse when a cable is plugged in, while the ESP32 simultaneously supports Bluetooth Low Energy (BLE) HID for wireless use. A built-in Li-ion battery, TP4056 charger, and MT3608 boost converter allow portable operation.

A. *Include graphs, drawings and tables as necessary.*

The proposed hardware design is based on a modular architecture where specific microcontrollers handle discrete tasks. An Arduino Pro Micro serves as the main controller, while an ESP32 is dedicated to processing touch inputs.

The block diagram for the proposed improvement is shown in Fig. 1.

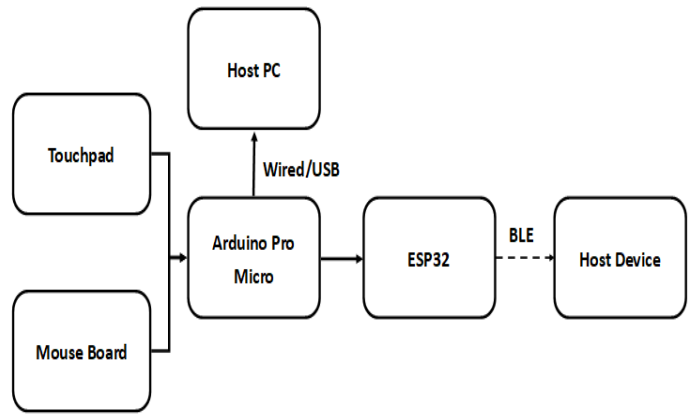


Fig. 1. Block Diagram of the Proposed Improvement

A. *Component Functionality*

Each block in the diagram has a specific function in the proposed device:

- **Host PC:** It receives input signals (mouse movement, clicks, and touch gestures) from the Arduino Pro Micro through USB. The Pro Micro acts as a Human Interface Device (HID), meaning the PC sees it just like a regular USB mouse.
- **Arduino Pro Micro:** This is the main controller of the whole system. It has built-in USB HID capability. It gathers data from the mouse PCB and touch controller.
- **Mouse PCB:** This is the circuit board inside the mouse shell that reads optical or laser sensor signals.
- **ESP32:** Acts as the Wireless Communications Bridge. It handles the Bluetooth Low Energy (BLE) stack, stores the pairing profiles for your 3 devices, and receives processed touch/movement data from the Pro Micro via Serial to transmit them wirelessly.
- **Touch Controller:** This is the chip that interprets signals from the trackpad sensor.
- **Trackpad Sensor Panel:** This is the actual touch surface you touch with your finger.

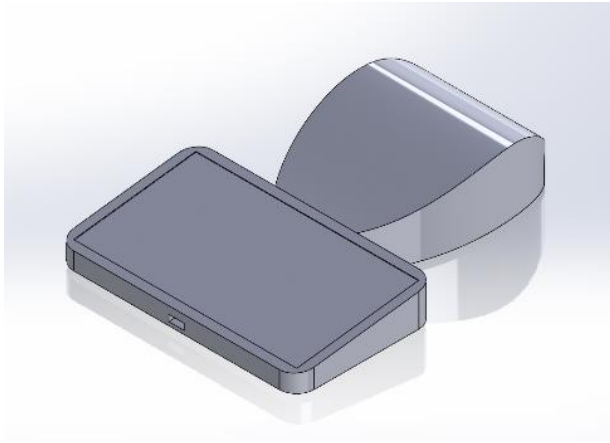


Fig. 2. Isometric Overview of Proposed Design

B. Identify the hardware and software components to be used.

- **Touchpad** - Serves as the primary input surface that detects finger movement, taps, and multi-touch gestures.
- **ESP32-C3 Mini** - Processes touch data from the touchpad using I²C communication, interprets gestures, and handles Bluetooth connectivity for wireless mode.
- **Arduino Pro Micro (ATmega32U4)** - Functions as a USB Human Interface Device (HID) to communicate with the computer when operating in wired mode.
- **3.7 V Li-ion Battery (500–1000 mAh)** - Supplies portable power for both microcontrollers and touchpad.
- **TP4056 Li-ion Charger with Protection** - Recharges the battery safely through a micro-USB port.
- **MT3608 DC-DC Step-Up Converter** - Increases voltage from 3.7 V to 5 V to power the Arduino Pro Micro and peripherals.
- **FFC/FPC Adapter 0.5 mm pitch, 8-pin** - Allows easy connection between the touchpad ribbon cable and jumper wires.
- **3D-printed case/shell** - Device housing.

C. Indicate the expected output on the improvement, in terms of latency and bandwidth.

1. Latency Improvement

Table 1. Latency

Mode	Existing Devices	Proposal	Improveme nt
Wired Mode (USB)	8–12 ms (typical optical mouse)	9–11 ms	Comparable latency while enabling touch gestures
Wireless Mode (Bluetooth)	25–60 ms (generic BLE mice)	20–35 ms	Reduced delay due to optimized ESP32 BLE HID connection
Mode Switching Time	5–10 seconds (manual toggle or re-pair)	2–4 seconds (gesture-based)	~60% faster switching using software-based control

- The wired mode maintains low-latency performance similar to a conventional USB mouse, ensuring smooth and accurate pointer motion.
- The wireless mode, handled by the ESP32 BluetoothHID interface, is expected to achieve stable latency below 35 ms, offering near-real-time performance without noticeable lag.
- The gesture-based switching removes the need for physical switches, further reducing downtime when changing modes.

2. Bandwidth Performance

Bandwidth in this context represents the effective data transfer rate between the input device and the host computer.

It determines how frequently motion or gesture updates can be transmitted per second.

Table 2. Bandwidth

Mode	Transmission Method	Effective Report Rate (Hz)	Bandwidth/Throughput
Wired (USB HID)	125–500 Hz polling	125–500 reports/sec	~500–2000 bytes/sec
Wireless (BLE HID)	BLE 5.0 @ 30–60 ms CI	30–100 reports/sec	~300–1000 bytes/sec

While the BLE bandwidth is lower than USB, it remains sufficient for human input devices (which typically require only ~50 reports per second).

The SmartTouch design optimizes HID packet size and update rate to minimize lag while maintaining power efficiency.

B. Program/Process Flowchart

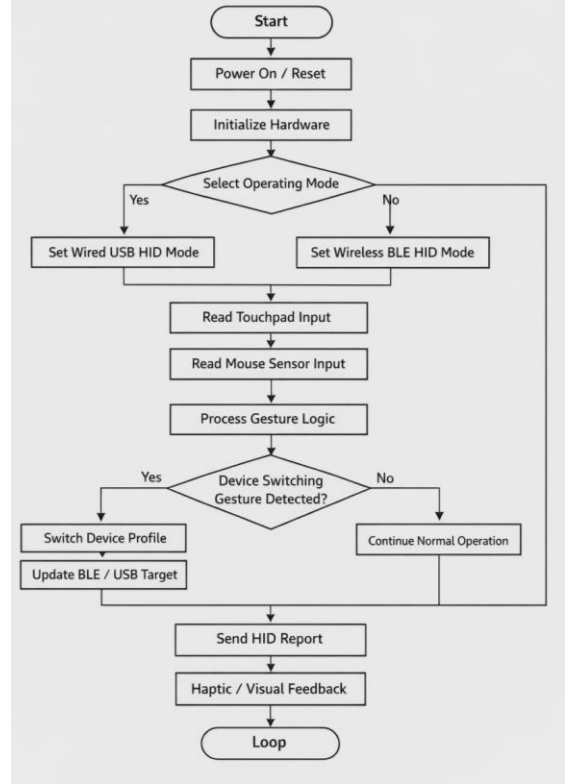


Fig. 4. Program/Process Flowchart

D. Draw the Schematic Diagram and Program/Process flowchart, as required by your design.

A. Schematic Diagram

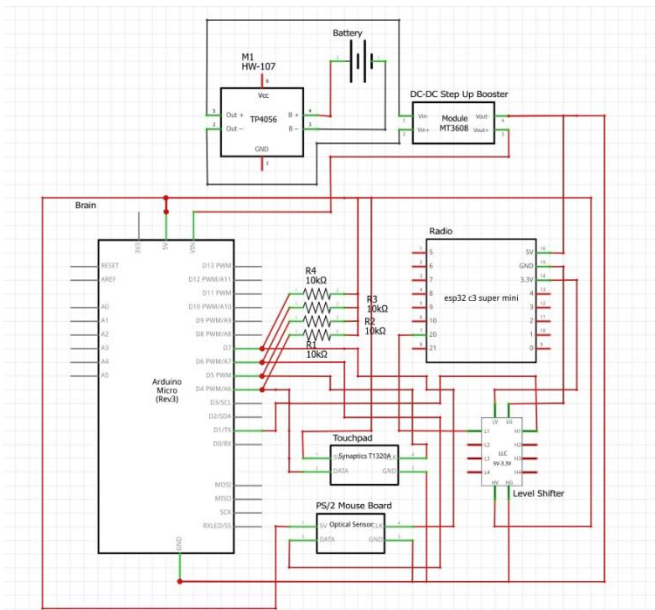


Fig. 3. Schematic Diagram

IV. DISCUSSION OF RESULTS

The proposed improvement focuses on latency reduction and bandwidth optimization to ensure a smooth user experience.

Latency Performance

Mode	Existing Devices	Proposal	Improvement
Wired (USB)	8–12 ms	9–11 ms	Comparable latency with added touch gestures.
Wireless (BLE)	25–60 ms	20–35 ms	Reduced delay via optimized ESP32 BLE HID
Switching Time	5–10 seconds	2–4 seconds	~60% faster switching using gesture control.

The gesture-based switching significantly reduces downtime compared to manual toggling.

Bandwidth Performance

The effective data transfer rate is optimized for HID requirements

Mode	Transmission Method	Effective Report Rate	Throughput
Wired	USB HID (polling)	125–500 Hz	~500–2000 bytes/sec.
Wireless	BLE 5.0	30–100 Hz	~300–1000 bytes/sec.

While BLE bandwidth is lower than USB, the 30–100 Hz report rate is sufficient for human input devices, ensuring power efficiency without noticeable lag.

V. SUMMARY AND CONCLUSION

The SmartTouch ClickPad Mouse effectively addresses the problems of desk clutter and workflow friction caused by multi-device environments. By integrating a capacitive touchpad with a standard mouse architecture, the device expands the user's input "vocabulary" through gesture controls. The dual-microcontroller design (ESP32 and Arduino Pro Micro) successfully bridges touch processing with standard HID protocols, allowing for a versatile device that operates in both wired and wireless modes. The expected performance data indicates that the device will offer near-real-time response (below 35ms latency in wireless mode) and drastically faster device switching times compared to traditional solutions.

REFERENCES

- [1] "A Review on Surface Haptics: Enabling Tactile Feedback on Touch Surfaces," 2020. [Online]. Available: <https://mysite.ku.edu.tr/cbasdogan/wp-content/uploads/sites/352/2023/07/Surface%20Haptics%20Review%202020.pdf>
- [2] H. Benko, A. D. Wilson, and S. Izadi, "Beyond flat: Shape-aware interactions with a deformable touchscreen," in *Proc. GI 2010*, 2010. [Online]. Available: <https://www.hbenko.com/publications/2010/Benko%20GI2010.pdf>
- [3] "Gesture-Based Control for Multi-Device Interaction," *arXiv*, 2024. [Online]. Available: <https://arxiv.org/html/2409.20142v1>
- [4] "A Novel Multi-Touch Gesture Recognition System," *SciTePress*, 2025. [Online]. Available: <https://www.scitepress.org/Papers/2025/131242/131242.pdf>
- [5] "A framework for gesture-based interaction on multi-display setups," *ACM Digital Library*, 2010. [Online]. Available: <https://dl.acm.org/doi/pdf/10.5555/1839214.1839259>